DT

DAVID K. TAN

(404) 452-8263
david.k.tan2@gmail.com
Website: Https://davidktan.com

CLEARANCE

TS/SAP Fully Adjudicated | Active Secret Clearance

TECHNICAL SKILLS

- Modeling & Simulation: 6-DOF Flight Dynamics, Missile Simulation, and Monte Carlo Analysis.
- Programming: C++, Python, Java.
- DevOps & Tooling: GitLab CI/CD, Jenkins and Docker.
- Guidance & Control: Autopilot Implementation and Control Logic Integration.
- Software Engineering: Object-Oriented Design (OOD), Agile Development.
- Systems & Environment: Linux/Unix, Red Hat Enterprise Linux (RHEL).

PROFESSIONAL EXPERIENCE

6-DOF Modeling & Simulation Engineer, **I3**, Mar 2026 - Current

- Developed and validated high-fidelity 6-DOF missile simulation models (C++), including implementation of autopilot control logic for LRPM and integration of guidance, sensor, and flight dynamics components across programs including Hydra-70 and APKWS; increased simulation throughput by 30%, reducing validation cycle time and enabling faster performance assessment.
- Executed verification and validation (V&V) simulation studies, translating requirements into testable scenarios and running Monte Carlo analyses to evaluate system performance; identified model discrepancies and improved accuracy, reducing rework and increasing confidence in mission-critical results.
- Collaborated with cross-functional engineering teams to refine model assumptions, troubleshoot simulation issues, and align results with system requirements; improved model reliability and reduced debugging/iteration time, supporting more efficient analysis cycles.

Software Engineer, **ALKU**, Sep 2024 - Jan 2026, Hill AFB, UT

- Engineered high-fidelity 6-DOF missile flight dynamics simulations (C++, Python) in support of U.S. Air Force programs including the LGM-35A Sentinel program, applying numerical integration and sensor modeling to increase trajectory prediction accuracy by 25% and strengthen confidence in performance assessments.
- Diagnosed and resolved simulation inconsistencies by analyzing flight behavior and refining underlying model assumptions, improving model fidelity and reducing downstream validation effort.
- Streamlined integration and release processes by implementing GitLab CI/CD pipelines (YAML), enhancing build reliability and accelerating iteration cycles for simulation updates.

Software Engineer II, **Space Dynamics Laboratory**, Jan 2023 - Sep 2024, Logan, UT

- Implemented high-fidelity 6-DOF missile simulation components (C++, Python) within existing flight dynamics frameworks, contributing to U.S. Air Force programs including the FORGE program and supporting trajectory prediction improvements of up to 25% through physics-based modeling and numerical methods.
- Supported integration and verification of simulation models, analyzing outputs to identify inconsistencies and assist in refining system behavior, improving overall model stability and reliability.

Computer Scientist Intern, **Bowie State University**, Jun 2022 - Aug 2022, Remote

- Led cybersecurity research within a distributed systems REU program by designing and executing simulated network attacks, improving threat detection effectiveness by 28% through enhanced traffic analysis and anomaly modeling.

PROJECT

Mission Planning Dashboard

- Built full-stack mission planning dashboard with role-based workflows, REST API endpoints, and audit logging, reducing planning data entry errors by 30%.

EDUCATION

Master of Science, Computer Science — Jul 2027
Georgia Institute of Technology - Atlanta, GA

MBA, Master of Business Administration — Dec 2023
Louisiana State University - Shreveport, LA

Bachelor of Science, Computer Science — Dec 2022
Valdosta State University - Valdosta, GA

CERTIFICATIONS

CompTIA Security+ | AWS Solutions Architect | CKA | Google Cloud Architect | CSEP

TECHNICAL LEADERSHIP

- Supported cross-functional integration between software, simulation, and mission operations teams in classified defense programs, improving coordination across model development, test planning, and mission analysis workflows.